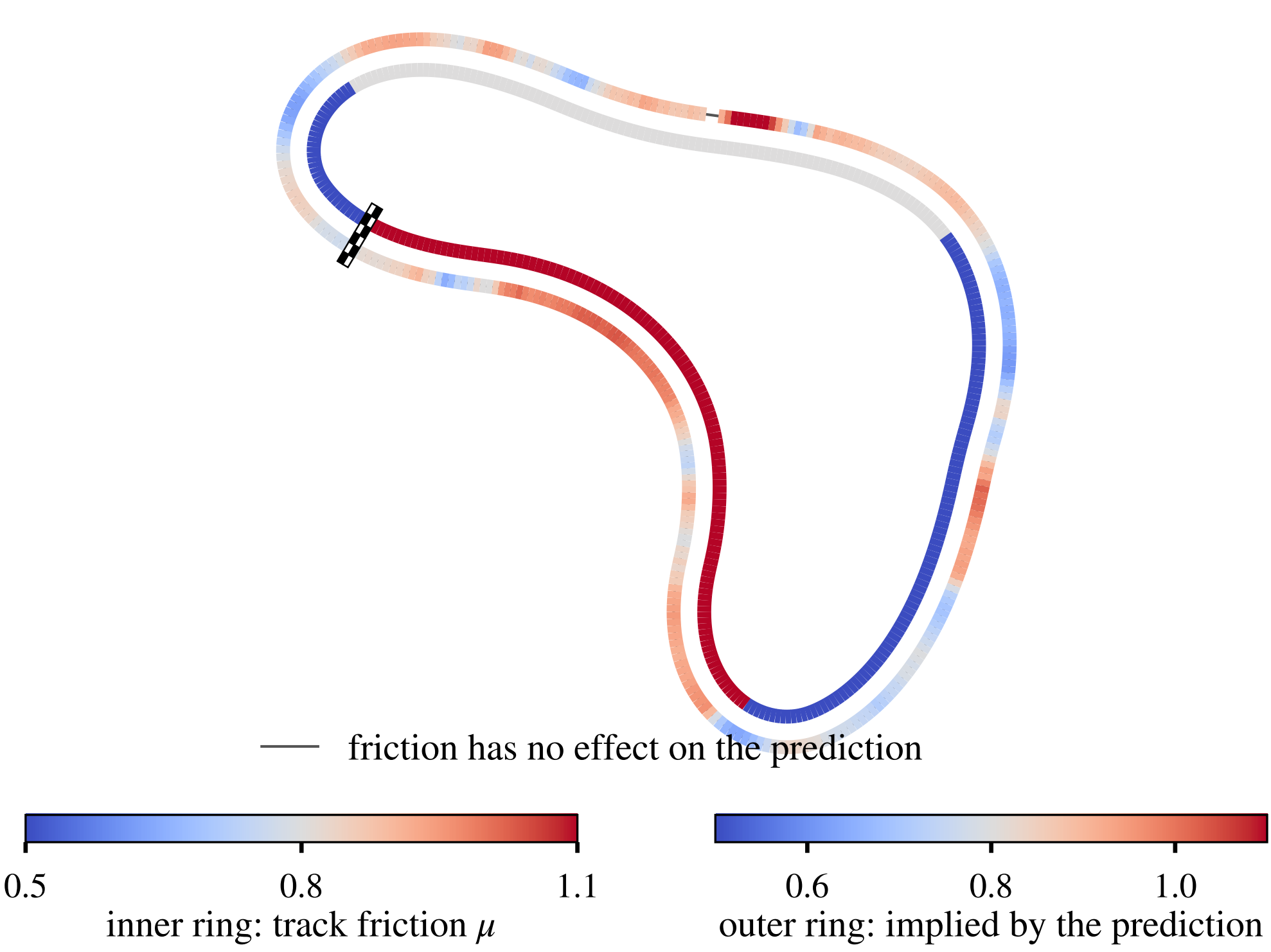
friction has no effect on the prediction
0.5
0.8
1.1
inner ring: track friction $\mu$
0.6
0.8
1.0
outer ring: implied by the prediction